MILITARY SPECIFICATION

INSULATION FELT, THERMAL, REFRACTORY FIBER, FLEXIBLE

1. SCOPE

1.1 Scope. - This specification covers flexible refractory fiber felt insulation for thermal control at temperatures up to 2000°F.

1.2 Grades and classes. - Insulation felt shall be of the following grades and classes, as specified (see 6.2):

Grade A - Temperature up to 1200°F.
  Class 1 - 3 pounds per cubic foot, density
  Class 2 - 6 pounds per cubic foot, density
Grade B - Temperatures up to 2000°F.
  Class 1 - 3 pounds per cubic foot, density
  Class 2 - 4 pounds per cubic foot, density
  Class 3 - 6 pounds per cubic foot, density

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids, forms a part of this specification:

SPECIFICATION

MILITARY
  MIL-P-116 - Preservation, Methods of.

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
  ASTM C-167-50 - Thickness and Density of Blanket or Batt-Type Thermal Insulation Materials.
  ASTM C-177-45 - Thermal Conductivity of Materials by Means of the Guarded Hotplate, Method of Test for.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

OFFICIAL CLASSIFICATION COMMITTEE
  Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material. - The insulation shall be composed of refined refractory fibers with or without binders to form a flexible felt.

3.2 Dimensions and densities. -

3.2.1 Dimensions. - The felt shall be furnished in the following nominal dimensions, as specified (see 6.2):

	Inches, nominal
Thickness	1/4, 3/8, 1/2, 3/4, 1
Width	24, 42
Length	36, 48, 96, 288

3.2.1.1 Tolerances. - A minus tolerance of 1/8-inch in width and length, and zero inch for thickness will be permitted. An excess in all dimensions will be permitted.

3.2.2 Density. - A minus tolerance of 5 percent and a plus tolerance of 20 percent of the densities shown in 1.2 will be permitted.

3.3 Physical requirements. - The insulation shall conform to the physical requirements shown in table I.

3.4 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Inspection lot. - For purposes of sampling, a lot shall consist of all felt of the same grade, class,

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width, length and thickness produced in one plant under essentially the same conditions and offered for delivery at one time.

4.2.2 Sampling for visual and dimensional examination. - Sample sheets or rolls shall be selected from each lot of insulation for the examination specified in 4.3.1, with lot acceptance based on table II.

4.2.3 Sampling for lot acceptance tests. - From each lot, 10 pieces 24 by 36 inches shall be selected at random for the test specified in 4.3.2. Each specimen shall be taken from a different sheet or roll.

4.3 Examination and tests. -

4.3.1 Examination. - Each of the sample sheets or rolls selected in accordance with 4.2.2 shall be surface examined, weighed, and measured to determine conformance with the requirements of this specification which do not require tests. Any sheet or roll in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sheets or rolls in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

Table I - Physical requirements

Characteristics	Grade A Classes		Grade B Classes		
	1	2	1	2	3
Compressive strength, to obtain nominal thickness, pounds per square feet (PSF)					
Minimum	0	0	0	0	0
Maximum	50	50	50	50	50
Tensile strength, pounds per square inch PSI (minimum)	1.0	1.0	0.2	0.3	0.4
Flexibility			no visible rupture		
Fiber diameter, inches $X10^{-4}$ (maximum)	1.5	1.5	3.5	3.5	3.5
Thermal conductivity, B.t.u. - in/ sq/ ft./hr/°F. temperature difference (maximum) at mean temperature of:					
400°F.	0.38	0.32	0.55	0.50	0.45
600°F.	0.55	0.45	0.80	0.80	0.55
900°F.	---	---	1.25	1.05	0.85
Changes under soaking heat, 24 hours at indicated temperature °F.	1200	1200	2000	2000	2000
Tensile strength, p.s.i. (minimum)	1.0	1.0	0.1	0.2	0.3
Loss in weight, percent (maximum)	1.0	1.0	2.0	2.0	2.0
Linear shrinkage, percent (maximum)	1.0	1.0	2.5	2.5	2.5
Deterioration of fiber, percent (maximum)	5	5	30	30	30

Table II - Sampling for visual and dimensional examination.

Number of sheets or rolls in inspection lot	Sample number of sheets or rolls	Acceptance number (defectives)	Rejection number (defectives)
15 and under	3	0	1
16 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
Over 110	25	2	3

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4.3.2 Lot acceptance tests. - The samples selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.4 to determine conformance with this specification except that the thermal conductivity test (see 4.4.8) shall be conducted only when specified (see 6.2).

4.3.2.1 Action in case of failure. - If any of the samples tested is found to be not in conformance with this specification, the lot which it represents shall be rejected.

4.4 Test procedures. -

4.4.1 Conditioning samples. - Specimens for test shall be dried to constant weight by heating to 220° to 230°F.

4.4.2 Thickness. - Test specimens, 12 inches square, shall be ruled off into four approximately square and equal areas 6 by 6 inches, and the thickness measurements taken at the center of each area. In determining the thickness the test specimen shall be placed on a hard, flat surface and covered with a 1/8-inch thick rigid loading plate weighing 1-1/4 PSF. The penetrating pin of a depth gage (see figure 1) shall be forced downward through oversized holes in the loading plate through the insulation to the hard flat surface. The sliding disk of the gage shall be lowered to the point of contact with the top surface of the loading plate. The gage shall be withdrawn, and the distance from the point of the pin to the sliding disk shall be measured to the nearest 1/32 inch. The actual thickness of the loading plate shall be subtracted from the overall pin to disk measurement and the average of twelve such determinations shall be taken as the thickness of the test specimen.

4.4.3 Compression load. - The load required to compress the insulation to nominal thickness shall be determined by either a dead weight or by mechanical loading. The weight shall be added or applied to the center of the loading plate (6 by 6 by 1/8 inch thick and weighing 1-1/4 PSF), and progressively increased until the insulation is compressed to its nominal thickness. The weight added per square foot shall be taken as the compression load to obtain nominal thickness.

4.4.4 Density. - The density of this insulation shall be determined in accordance with the method specified in ASTM Standard C167-50, except that the thickness measurement as determined in 4.4.2 and the nominal thickness measurement as determined in 4.4.3 shall be used to calculate the density.

4.4.5 Tensile strength. - The tensile strength of the insulation shall be determined with 12-inch wide and 14-inch long specimens. The specimens shall be suspended vertically and held with two clamping devices positioned 12 inches apart. The clamps shall be formed by bolting together at the ends two 14-inch lengths of 1/2 inch pipe. Sponge rubber, 1/4 inch thick, shall be cemented to the outer surface of the pipe in the clamping area to prevent cutting of the specimens. The clamped specimens shall be pulled to rupture attached to the jaws of a tensile testing machine, the lower jaw of which moves at a uniform rate of 1 plus or minus 0.1 inch per minute. The tensile strength shall be taken as the average of three determinations and shall be expressed in PSI of cross sectional area using nominal thickness in the computation.

4.4.6 Flexibility. - Specimens of insulation, 12 by 12 inches by specified thickness, shall be bent over an angle of 90 degrees and examined for rupture on the outside surface. Any minor surface separations shall be disregarded. The radii over which the material shall be bent, shall be as follows:

Grade	Class	Radius, inches
A	1	2
	2	6
B	1	2
	2	4
	3	6

4.4.7 Fineness of fiber. - The diameter of individual fibers shall be determined microscopically on the basis of measuring ten consecutive fibers from each of five clusters selected at random from the insulation. The average of all measurements shall be reported as the fiber diameter.

4.4.8 Thermal conductivity. - Thermal conductivity shall be determined in accordance with ASTM Method C177-45.

4.4.9 Deterioration of fibers. - Ten-gram samples composed of 5/8-inch diameter plugs cut from the insulation, as with a cork borer, shall be used in the fiber deterioration test. The samples shall be weighed, placed in a covered 8-inch diameter 25 mesh sieve over a retaining pan secured in a standard laboratory testing sieve on a shaker operated with a horizontal eccentric circular motion at 285 plus or minus 10 revolutions per minute and a tapping action of 150 plus or minus 5 strokes per minute. After shaking 10 minutes with the tapper in operation, the material collected in the retaining pan shall be weighed and the percentage of the total sample calculated. The difference between values obtained with samples of the insulation in the as received condition and those after the soaking heat tests shall be used to express the fiber deterioration of the insulation.

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4.4.10 Physical changes under soaking heat.- Specimens shall be weighed and measured. Then the specimens shall be placed in an electrically heated oven and subjected to the maximum temperature for the respective grade and class for 24 hours. The specimens shall be removed from the oven and tested to determine conformance with the requirements specified in table I.

4.5 Examination of preparation for delivery.- Sample felts and shipping containers shall be selected and examined in accordance with MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.-

5.1.1 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.2 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Packing.-

5.2.1.1.1 Level A.- Insulation felt shall be packed in a container conforming to PPP-P-636, class 2. All corners and edge seams, and manufacturer's joint shall be waterproofed in accordance with the appendix to PPP-B-636.

5.2.1.1.2 Level B.- Insulation felt shall be packed in fiberboard boxes conforming to class 1 of PPP-B-636.

5.2.2 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use.- Grade B material is intended for use for thermal control of machinery and equipment such as steam or gas tubines, boilers, and piping at temperatures up to 2000°F. Grade A material is intended as a lower-temperature outer layer component for grade B material.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of specification.
(b) Grade and class required (see 1.2).
(c) Thickness, length and width required (see 3.2).
(d) When thermal conductivity tests should be conducted (see 4.3.2).
(e) Packing or marking requirements other than required by 5.1) (see 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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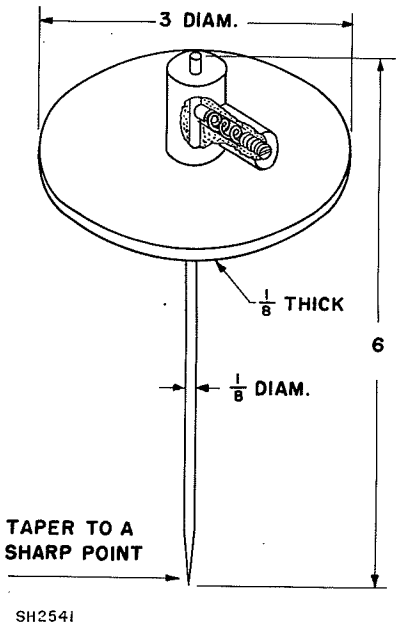


Figure 1 - Dept gage for thickness measurements.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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